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Expression of beta-glucosidase in yeast for improved ethanol production

Abstract

Described are compositions and methods relating to the expression of β -glucosidase in yeast for use in industrial ethanol production. The yeast demonstrates increased ethanol production and reduced production of unwanted side-products, including acetate. Such yeast is particularly useful for large-scale ethanol production from starch substrates.

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Background/Summary

TECHNICAL FIELD

(1) The present compositions and methods relate to the expression of β -glucosidase in yeast for use in industrial ethanol production. The yeast demonstrates increased ethanol production and reduced production of unwanted side-products, including acetate. Such yeast is particularly useful for large-scale ethanol production from starch substrates.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

(2) The official copy of the sequence listing is submitted electronically via EFS-Web as an ASCII formatted sequence listing with a file named "20190829_NB41307USPSP_SeqLst.txt created on Aug. 29, 2019 and having a size of 62 of kbs kilobytes and is filed concurrently with the specification. The sequence listing contained in this ASCII formatted document is part of the specification and is herein incorporated by reference in its entirety.

BACKGROUND

(3) First-generation yeast-based ethanol production converts sugars into fuel ethanol. The annual fuel ethanol production by yeast is about 90 billion liters worldwide (Gombert, A. K. and van Maris, A. J. (2015) *Curr. Opin. Biotechnol.* 33:81-86). It is estimated that about 70% of the cost of ethanol production is the feedstock. Since the production volume is so large, even small yield improvements have massive economic impact across the industry.

(4) The Renewable Fuel Standard (RFS) is a federal program that requires a minimum volume of renewable fuels to be blended into transportation fuel sold in the United States. The RFS originated with the Energy Policy Act of 2005 and was extended and expanded in the Energy Independence and Security Act of 2007. In 2010 the Environmental Protection Agency (EPA) established a process for companies to petition for new fuel pathways to qualify for the (RFS) program. A fuel pathway is a specific combination of (1) a feedstock, (2) a production process and (3) a fuel type, wherein each combination of three components represents a separate fuel pathway. Qualifying fuel pathways are assigned one or more D-codes corresponding to the type of Renewable Identification Number (RIN) they are eligible to generate. Conventional renewable fuel (e.g., from corn) is D6, advanced biofuel is D5, biodiesel is D4 and cellulosic biofuel is D3 or D7. Cellulosic Biofuel (D-Codes 3 and 7) must be produced from cellulose, hemicellulose or lignin.

(5) RINs are tradable regulatory credits that represent a quantity of qualifying renewable fuel. RINs are assigned after a producer reports the production of a gallon of fuel to the EPA. Blenders demonstrate compliance with the RFS by turning RINs over to the EPA once the gallon of fuel is blended into transportation fuel. Because the RFS requires increasing amounts of advanced biofuels (including cellulosic biofuels) as time progresses, RINs have different values depending on the fuel pathway from which they are generated. For example, a D3 RIN is currently worth more than a D6 RIN.

(6) The current National Renewable Energy Laboratory (NREL) laboratory analytical procedure (LAP) for determination of structural carbohydrates and lignin in biomass is described by Sluiter, A. et al. ((2008) NREL Laboratory Analytical Procedure NREL/TP-510-42618. Golden, CO: National Renewable Energy Laboratory). The method is based on two-step acid hydrolysis, in which biomass is first hydrolyzed using 72 wt % sulfuric acid at 30° C. for 1 h, followed by dilution to 4 wt % sulfuric acid for further hydrolysis at 121° C. for 1 h under autoclave conditions. A faster, single-step method has been described in which biomass is hydrolyzed in 4 wt % sulfuric acid for further hydrolysis at 121° C. for 1 h under autoclave conditions (Gao, X. et al. (2014) *Biotechnology and Bioengineering* 111:1088-96). In both methods, the products of hydrolysis are analyzed by HPLC. A limitation of these methods is that they cannot distinguish glucose derived from starch as opposed to glucose derived from cellulose and, therefore cannot be used to determine the fraction of glucose derived from cellulose in a mixed starch-cellulose feedstock. The EPA established corn kernel fiber as a qualified crop residue on Jul. 18, 2014.

(7) As corn ethanol producers attempt to utilize corn fiber, as well as corn starch, to produce

ethanol, there is a financial incentive to characterize as much ethanol as possible as D3 biofuel. However, the EPA requires accuracy in accounting and producers that have non-accurately characterized their biofuel can be subject to penalties. Accordingly, the need exists for an accurate method for determining the source of ethanol when mixed feedstocks of starch and cellulosic components are used to produce biofuels.

SUMMARY

(8) The present compositions and methods relate to modified yeast that expresses β -glucosidase. Aspects and embodiments of the compositions and methods are described in the following, independently-numbered, paragraphs.

(9) 1. In one aspect, modified yeast cells derived from parental yeast cells are provided, the modified cells comprising a genetic alteration that causes the modified cells to produce an increased amount of β -glucosidase polypeptides compared to the parental cells, wherein the modified cells produce during fermentation more ethanol and/or less acetate compared to the amount of ethanol and acetate produced by otherwise identical parental yeast cells.

(10) 2. In some embodiments of the modified cells of paragraph 1, the genetic alteration comprises the introduction into the parental cells of a nucleic acid capable of directing the expression of a β -glucosidase polypeptide to a level above that of the parental cell grown under equivalent conditions.

(11) 3. In some embodiments of the modified cells of paragraph 1, the genetic alteration comprises the introduction of an expression cassette for expressing a β -glucosidase polypeptide.

(12) 4. In some embodiments of the modified cells of any of paragraphs 1-3, the cells further comprise an exogenous gene encoding a carbohydrate processing enzyme.

(13) 5. In some embodiments, the modified cells of any of paragraphs 1-4 further comprise a PKL pathway.

(14) 6. In some embodiments, the modified cells of any of paragraphs 1-5 further comprise an alteration in the glycerol pathway and/or the acetyl-CoA pathway.

(15) 7. In some embodiments, the modified cells of any of paragraphs 1-6 further comprise an alternative pathway for making ethanol.

(16) 8. In some embodiments of the modified cells of any of paragraphs 1-7, the modified cells further make reduced amounts of DP2 and/or DP3 compared to otherwise identical parental cells.

(17) 9. In some embodiments of the modified cells of any of paragraphs 1-8, the β -glucosidase polypeptides are derived from *Glomerella graminicola*.

(18) 10. In some embodiments of the modified cells of any of paragraphs 1-9, the β -glucosidase polypeptides have: (a) the amino acid sequence of SEQ ID NO: 1 or SEQ ID NO: 2, (b) an amino acid sequence having at least 80% amino acid sequence identity to SEQ ID NO: 1 or SEQ ID NO: 2, (c) the amino acid sequence of a mature polypeptide encoded by any of SEQ ID NOs: 11-16, (d) the amino acid sequence of a mature polypeptide encoded by a nucleic acid having at least 80% nucleic acid identity to any of SEQ ID NOs: 11-16, or (e) the amino acid sequence of a mature polypeptide encoded by a nucleic acid that hybridizes, under stringent condition, to any of SEQ ID NOs: 11-16, or the complement, thereof.

(19) 11. In some embodiments of the modified cells of any of paragraphs 1-10, the cells are of a *Saccharomyces* spp.

(20) 12. In another aspect, a method for increasing the production of alcohol and/or decreasing the production of acetate from yeast cells grown on a carbohydrate substrate is provided, comprising: introducing into parental yeast cells a genetic alteration that increases the production of β -glucosidase polypeptides compared to the amount produced in the parental cells.

(21) 13. In some embodiments of the method of paragraph 12, the cells having the introduced genetic alteration are the modified cells are the cells of any of paragraphs 1-11.

(22) 14. In some embodiments of the method of paragraph 12 or 13, the increased production of alcohol is at least 0.5%, at least 1.0%, at least 2.0% or at least 3.0%.

(23) 15. In some embodiments of the method of any of paragraphs 12-14, the decreased production of acetate is at least 1.0%, at least 2.0%, at least 4.0% or at least 6.0%.

(24) 16. In some embodiments of the method of any of paragraphs 12-15, the cells having the introduced genetic alteration comprise an exogenous PKL pathway.

(25) These and other aspects and embodiments of present modified cells and methods will be apparent from the description, including any accompanying Drawings/Figures.

BRIEF DESCRIPTION OF THE SEQUENCES

(26) SEQ ID NO: 1 represents the predicted mature amino acid sequence of ABG54.

(27) SEQ ID NO: 2 represents the predicted mature amino acid sequence of FAB, which is the same as SEQ ID NO: 135 of WO2012125951 (Kaper et al.).

(28) SEQ ID NO: 3 represents the predicted mature amino acid sequence of Mg3A.

(29) SEQ ID NO: 4 represents the predicted mature amino acid sequence of TrBGL1.

(30) SEQ ID NO: 5 represents the predicted native signal sequence for ABG54.

(31) SEQ ID NO: 6 represents the predicted native signal sequence for FAB.

(32) SEQ ID NO: 7 represents the predicted native signal sequence for Mg3A.

(33) SEQ ID NO: 8 represents the predicted native signal sequence for TrBGL1.

(34) SEQ ID NO: 9 represents the predicted MFalpha signal sequence.

(35) SEQ ID NO: 10 represents the predicted SUC2 signal sequence.

(36) SEQ ID NO: 11 represents the nucleotide sequence encoding native-ABG54 in pYKH1127, GKH-0464, G3020, G3014, GKH-0737 and GKH-0732.

(37) SEQ ID NO: 12 represents the nucleotide sequence encoding MFalpha-ABG54 in pYKH1139 and GKH-0459.

(38) SEQ ID NO: 13 represents the nucleotide sequence encoding SUC2-ABG54 in GKH-0484.

(39) SEQ ID NO: 14 represents the nucleotide sequence encoding native-FAB in pYKH1135 and GKH-0455.

(40) SEQ ID NO: 15 represents the nucleotide sequence encoding MFalpha-FAB in pYKH1095 and GKH-0450.

(41) SEQ ID NO: 16 represents the nucleotide sequence encoding SUC2-FAB in GKH-0466.

(42) SEQ ID NO: 17 represents the nucleotide sequence encoding native-Mg3A in pYKH1097.

(43) SEQ ID NO: 18 represents the nucleotide sequence encoding MFalpha-Mg3A in pYKH1096.

(44) SEQ ID NO: 19 represents the nucleotide sequence encoding native-TrBGL1 in pYKH1099.

(45) SEQ ID NO: 20 represents the nucleotide sequence encoding MFalpha-TrBGL1 in pYKH1098.

Description

DETAILED DESCRIPTION

I. Definitions

(1) Prior to describing the present yeast and methods in detail, the following terms are defined for clarity. Terms not defined should be accorded their ordinary meanings as used in the relevant art.

(2) As used herein, the term “alcohol” refers to an organic compound in which a hydroxyl functional group (—OH) is bound to a saturated carbon atom.

(3) As used herein, the terms “yeast cells,” “yeast strains,” or simply “yeast” refer to organisms from the phyla Ascomycota and Basidiomycota. Exemplary yeast is budding yeast from the order Saccharomycetales. Particular examples of yeast are *Saccharomyces* spp., including but not limited to *S. cerevisiae*. Yeast include organisms used for the production of fuel alcohol as well as organisms used for the production of potable alcohol, including specialty and proprietary yeast strains used to make distinctive-tasting beers, wines, and other fermented beverages.

(4) As used herein, the phrase “engineered yeast cells,” “variant yeast cells,” “modified yeast

cells,” or similar phrases, refer to yeast that include genetic modifications and characteristics described herein. Variant/modified yeast do not include naturally occurring yeast.

(5) As used herein, the terms “polypeptide” and “protein” (and their respective plural forms) are used interchangeably to refer to polymers of any length comprising amino acid residues linked by peptide bonds. The conventional one-letter or three-letter codes for amino acid residues are used herein and all sequence are presented from an N-terminal to C-terminal direction. The polymer can comprise modified amino acids, and it can be interrupted by non-amino acids. The terms also encompass an amino acid polymer that has been modified naturally or by intervention; for example, disulfide bond formation, glycosylation, lipidation, acetylation, phosphorylation, or any other manipulation or modification, such as conjugation with a labeling component. Also included within the definition are, for example, polypeptides containing one or more analogs of an amino acid (including, for example, unnatural amino acids, etc.), as well as other modifications known in the art.

(6) As used herein, functionally and/or structurally similar proteins are considered to be “related proteins,” or “homologs.” Such proteins can be derived from organisms of different genera and/or species, or different classes of organisms (e.g., bacteria and fungi), or artificially designed. Related proteins also encompass homologs determined by primary sequence analysis, determined by secondary or tertiary structure analysis, or determined by immunological cross-reactivity, or determined by their functions.

(7) As used herein, the term “homologous protein” refers to a protein that has similar activity and/or structure to a reference protein. It is not intended that homologs necessarily be evolutionarily related. Thus, it is intended that the term encompass the same, similar, or corresponding enzyme(s) (i.e., in terms of structure and function) obtained from different organisms. In some embodiments, it is desirable to identify a homolog that has a quaternary, tertiary and/or primary structure similar to the reference protein. In some embodiments, homologous proteins induce similar immunological response(s) as a reference protein. In some embodiments, homologous proteins are engineered to produce enzymes with desired activity(ies).

(8) The degree of homology between sequences can be determined using any suitable method known in the art (see, e.g., Smith and Waterman (1981) *Adv. Appl. Math.* 2:482; Needleman and Wunsch (1970) *J. Mol. Biol.*, 48:443; Pearson and Lipman (1988) *Proc. Natl. Acad. Sci. USA* 85:2444; programs such as GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package (Genetics Computer Group, Madison, WI); and Devereux et al. (1984) *Nucleic Acids Res.* 12:387-95).

(9) For example, PILEUP is a useful program to determine sequence homology levels. PILEUP creates a multiple sequence alignment from a group of related sequences using progressive, pair-wise alignments. It can also plot a tree showing the clustering relationships used to create the alignment. PILEUP uses a simplification of the progressive alignment method of Feng and Doolittle, (Feng and Doolittle (1987) *J. Mol. Evol.* 35:351-60). The method is similar to that described by Higgins and Sharp ((1989) *CABIOS* 5:151-53). Useful PILEUP parameters including a default gap weight of 3.00, a default gap length weight of 0.10, and weighted end gaps. Another example of a useful algorithm is the BLAST algorithm, described by Altschul et al. ((1990) *J. Mol. Biol.* 215:403-10) and Karlin et al. ((1993) *Proc. Natl. Acad. Sci. USA* 90:5873-87). One particularly useful BLAST program is the WU-BLAST-2 program (see, e.g., Altschul et al. (1996) *Meth. Enzymol.* 266:460-80). Parameters “W,” “T,” and “X” determine the sensitivity and speed of the alignment. The BLAST program uses as defaults a word-length (W) of 11, the BLOSUM62 scoring matrix (see, e.g., Henikoff and Henikoff (1989) *Proc. Natl. Acad. Sci. USA* 89:10915) alignments (B) of 50, expectation (E) of 10, M⁵, N⁻⁴, and a comparison of both strands.

(10) As used herein, the phrases “substantially similar” and “substantially identical,” in the context of at least two nucleic acids or polypeptides, typically means that a polynucleotide or polypeptide comprises a sequence that has at least about 70% identity, at least about 75% identity, at least about

80% identity, at least about 85% identity, at least about 90% identity, at least about 91% identity, at least about 92% identity, at least about 93% identity, at least about 94% identity, at least about 95% identity, at least about 96% identity, at least about 97% identity, at least about 98% identity, or even at least about 99% identity, or more, compared to the reference (i.e., wild-type) sequence. Percent sequence identity is calculated using CLUSTAL W algorithm with default parameters. See Thompson et al. (1994) *Nucleic Acids Res.* 22:4673-4680. Default parameters for the CLUSTAL W algorithm are:

(11) TABLE-US-00001 Gap opening penalty: 10.0 Gap extension penalty: 0.05 Protein weight matrix: BLOSUM series DNA weight matrix: IUB Delay divergent sequences %: 40 Gap separation distance: 8 DNA transitions weight: 0.50 List hydrophilic residues: GPSNDQEKR Use negative matrix: OFF Toggle Residue specific penalties: ON Toggle hydrophilic penalties: ON Toggle end gap separation penalty OFF

(12) Another indication that two polypeptides are substantially identical is that the first polypeptide is immunologically cross-reactive with the second polypeptide. Typically, polypeptides that differ by conservative amino acid substitutions are immunologically cross-reactive. Thus, a polypeptide is substantially identical to a second polypeptide, for example, where the two peptides differ only by a conservative substitution. Another indication that two nucleic acid sequences are substantially identical is that the two molecules hybridize to each other under stringent conditions (e.g., within a range of medium to high stringency).

(13) As used herein, the term “hybridization” refers to the process by which one strand of nucleic acid forms a duplex with, i.e., base pairs with, a complementary strand, as occurs during blot hybridization techniques and PCR techniques. Stringent hybridization conditions are exemplified by hybridization under the following conditions: 65° C. and 0.1×SSC (where 1×SSC=0.15 M NaCl, 0.015 M Na₃ citrate, pH 7.0). Hybridized, duplex nucleic acids are characterized by a melting temperature (T_{sub.m}), where one-half of the hybridized nucleic acids are unpaired with the complementary strand. Mismatched nucleotides within the duplex lower the T_{sub.m}. A nucleic acid encoding a variant α-amylase may have a T_{sub.m} reduced by 1° C.-3° C. or more compared to a duplex formed between the nucleotide of SEQ ID NO: 2 and its identical complement.

(14) As used herein, the term “gene” is synonymous with the term “allele” in referring to a nucleic acid that encodes and directs the expression of a protein or RNA. Vegetative forms of filamentous fungi are generally haploid, therefore a single copy of a specified gene (i.e., a single allele) is sufficient to confer a specified phenotype. The term “allele” is generally preferred when an organism contains more than one similar genes, in which case each different similar gene is referred to as a distinct “allele.”

(15) As used herein, the term “expressing a polypeptide” and similar terms refers to the cellular process of producing a polypeptide using the translation machinery (e.g., ribosomes) of the cell.

(16) As used herein, an “expression cassette” refers to a DNA fragment that includes a promoter, and amino acid coding region and a terminator (i.e., promoter::amino acid coding region::terminator) and other nucleic acid sequence needed to allow the encoded polypeptide to be produced in a cell. Expression cassettes can be exogenous (i.e., introduced into a cell) or endogenous (i.e., extant in a cell).

(17) As used herein, the terms “wild-type” and “native” are used interchangeably and refer to genes, proteins or strains found in nature, or that are not intentionally modified for the advantage of the presently described yeast.

(18) As used herein, the term “protein of interest” refers to a polypeptide that is desired to be expressed in modified yeast. Such a protein can be an enzyme, a substrate-binding protein, a surface-active protein, a structural protein, a selectable marker, a signal transducer, a receptor, a transporter, a transcription factor, a translation factor, a co-factor, or the like, and can be expressed. The protein of interest is encoded by an endogenous gene or a heterologous gene (i.e., gene of interest) relative to the parental strain. The protein of interest can be expressed intracellularly or as

a secreted protein.

(19) As used herein, the terms “genetic manipulation” and “genetic alteration” are used interchangeably and refer to the alteration/change of a nucleic acid sequence. The alteration can include but is not limited to a substitution, deletion, insertion or chemical modification of at least one nucleic acid in the nucleic acid sequence.

(20) As used herein, a “functional polypeptide/protein” is a protein that possesses an activity, such as an enzymatic activity, a binding activity, a surface-active property, a signal transducer, a receptor, a transporter, a transcription factor, a translation factor, a co-factor, or the like, and which has not been mutagenized, truncated, or otherwise modified to abolish or reduce that activity. Functional polypeptides can be thermostable or thermolabile, as specified.

(21) As used herein, “a functional gene” is a gene capable of being used by cellular components to produce an active gene product, typically a protein. Functional genes are the antithesis of disrupted genes, which are modified such that they cannot be used by cellular components to produce an active gene product, or have a reduced ability to be used by cellular components to produce an active gene product.

(22) As used herein, yeast cells have been “modified to prevent the production of a specified protein” if they have been genetically or chemically altered to prevent the production of a functional protein/polypeptide that exhibits an activity characteristic of the wild-type protein. Such modifications include, but are not limited to, deletion or disruption of the gene encoding the protein (as described, herein), modification of the gene such that the encoded polypeptide lacks the aforementioned activity, modification of the gene to affect post-translational processing or stability, and combinations, thereof.

(23) As used herein, “aerobic fermentation” refers to growth in the presence of oxygen.

(24) As used herein, “anaerobic fermentation” refers to growth in the absence of oxygen.

(25) As used herein, the expression “end of fermentation” refers to the stage of fermentation when the economic advantage of continuing fermentation to produce a small amount of additional alcohol is exceeded by the cost of continuing fermentation in terms of fixed and variable costs. In a more general sense, “end of fermentation” refers to the point where a fermentation will no longer produce a significant amount of additional alcohol, i.e., no more than about 1% additional alcohol, or no more substrate left for further alcohol production.

(26) As used herein, the phrase “degree of polymerization” (DP) refers to the number of anhydroglucopyranose units in a given saccharide. An examples of DP1 is the monosaccharides glucose. Examples of DP2 are the disaccharides maltose and isomaltose.

(27) As used herein, the expression “carbon flux” refers to the rate of turnover of carbon molecules through a metabolic pathway. Carbon flux is regulated by enzymes involved in metabolic pathways, such as the pathway for glucose metabolism and the pathway for maltose metabolism.

(28) As used herein, the singular articles “a,” “an” and “the” encompass the plural referents unless the context clearly dictates otherwise. All references cited herein are hereby incorporated by reference in their entirety. The following abbreviations/acronyms have the following meanings unless otherwise specified: ° C. degrees Centigrade AA α -amylase AADH acetaldehyde dehydrogenases ADH alcohol dehydrogenase bp base pairs DNA deoxyribonucleic acid ds or DS dry solids DP degree of polymerization EC enzyme commission EtOH ethanol g or gm gram g/L grams per liter GA glucoamylase HPLC high performance liquid chromatography hr or h hour M molar mg milligram min minute mL or ml milliliter mM millimolar N normal n/a not applicable n/d no data nm nanometer PCR polymerase chain reaction PKL phosphoketolase ppm parts per million PTA phosphotransacetylase RPM revolutions per minute Δ relating to a deletion μ g microgram μ L and μ l microliter μ M micromolar

II. Modified Yeast Cells Expressing β -Glucosidase

(29) Described are modified yeast cells that express β -glucosidase, and methods of use, thereof. The yeast produces increased amounts of ethanol, decreased amounts of acetate, and offer

additional advantages compared to conventional yeast. The additional ethanol results from the utilization of non-starch-derived β -D-linked glucan substrates, for example, cellobiose and other cellooligosaccharide substrates, and may qualify for D3 RIN credits. Expression of β -glucosidase in yeast was inspired by the observation that contacting stillage with a cocktail of eight glycosyl hydrolase enzymes and enriched for β -glucosidase, liberated more glucose compared to cocktails enriched for a different glycosyl hydrolase.

(30) Yeast can be selective in their ability to express exogenous proteins, and expression of β -glucosidase is no exception. Expression of β -glucosidase in yeast was not straightforward and suitable yeast were not readily obtained. It is also surprising that the expression of β -glucosidase by an ethanologen, so far downstream of initial starch liquefaction, would lead to the observed benefits in ethanol and acetate production. Expression of β -glucosidase was also found to benefit ethanol and acetate production in combination with an exogenous PKL pathway and in combination with glucoamylase expression.

(31) In some embodiments, the increase in ethanol production by the modified cells is an increase of at least 0.5%, at least 0.7%, at least 0.9%, at least 1.2%, at least 1.5%, at least 2.0%, at least 3.0%, or more, compared to the amount of ethanol produced by parental cells grown under the same conditions.

(32) In some embodiments, the decrease in acetate production by the modified cells is a decrease of at least 0.5%, at least 1.0%, at least 2.0%, at least 3.0%, at least 4.0%, at least 5.0%, at least 6.0%, or more, compared to the amount of acetate produced by parental cells grown under the same conditions.

(33) Preferably, increased β -glucosidase expression is achieved by genetic manipulation using sequence-specific molecular biology techniques, as opposed to chemical mutagenesis, which is generally not targeted to specific nucleic acid sequences. However, chemical mutagenesis is not excluded as a method for making modified yeast cells.

(34) In some embodiments, the present compositions and methods involve introducing into yeast cells a nucleic acid capable of directing the expression, or over-expression, of a β -glucosidase polypeptide. Particular methods include but are not limited to (i) introducing an exogenous expression cassette for producing the polypeptide into a host cell, optionally in addition to an endogenous expression cassette, (ii) substituting an exogenous expression cassette with an endogenous cassette that allows the production of an increased amount of the polypeptide, (iii) modifying the promoter of an endogenous expression cassette to increase expression, (iv) increase copy number of the same or different cassettes for over-expression of β -glucosidase, and/or (v) modifying any aspect of the host cell to increase the half-life of the polypeptide in the host cell.

(35) In some embodiments, the parental cell that is modified already includes a gene of interest, such as a gene encoding a selectable marker, carbohydrate-processing enzyme, or other polypeptide. In some embodiments, a gene of interest is subsequently introduced into the modified cells.

(36) In some embodiments, the parental cell that is modified already includes an engineered pathway of interest, such as a PKL pathway to increase ethanol production, or any other pathway to increase alcohol production.

(37) In some embodiments of the present compositions and methods, the amino acid sequence of the β -glucosidase polypeptide that is expressed in modified yeast cells has at least about 80%, at least about 85%, at least about 87%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or even at least about 99% identity, to SEQ ID NO: 1 or SEQ ID NO: 2.

(38) In some embodiments of the present compositions and methods, β -glucosidase polypeptide has an the amino acid sequence of a mature polypeptide encoded by a nucleic acid having at least about 80%, at least about 85%, at least about 87%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about

97%, at least about 98%, or even at least about 99% identity nucleic acid identity to any of SEQ ID NOs: 11-16, or the amino acid sequence of a mature polypeptide encoded by a nucleic acid that hybridizes, under stringent condition, to any of SEQ ID NOs: 11-16, or the complement, thereof.

III. Modified Yeast Cells Having Increased β -Glucosidase Expression in Combination with Genes of an Exogenous PKL Pathway

(39) Increased expression of β -glucosidase can be combined with expression of genes in the PKL pathway to further increase ethanol production. Engineered yeast cells having a heterologous PKL pathway have been previously described in WO2015148272 (Miasnikov et al.). These cells express heterologous phosphoketolase (PKL), phosphotransacetylase (PTA) and acetylating acetyl dehydrogenase (AADH), optionally with other enzymes, to channel carbon flux away from the glycerol pathway and toward the synthesis of acetyl-CoA, which is then converted to ethanol. Such modified cells are capable of increased ethanol production in a fermentation process when compared to otherwise-identical parent yeast cells.

IV. Combination of Increased β -Glucosidase Production with Other Mutations that Affect Alcohol Production

(40) In some embodiments, in addition to expressing increased amounts of β -glucosidase polypeptides, optionally in combination with introducing an exogenous PKL pathway, the present modified yeast cells include additional beneficial modifications.

(41) The modified cells may further include mutations that result in attenuation of the native glycerol biosynthesis pathway and/or reuse glycerol pathway, which are known to increase alcohol production. Methods for attenuation of the glycerol biosynthesis pathway in yeast are known and include reduction or elimination of endogenous NAD-dependent glycerol 3-phosphate dehydrogenase (GPD) or glycerol phosphate phosphatase activity (GPP), for example by disruption of one or more of the genes GPD1, GPD2, GPP1 and/or GPP2. See, e.g., U.S. Pat. No. 9,175,270 (Elke et al.), U.S. Pat. No. 8,795,998 (Pronk et al.) and U.S. Pat. No. 8,956,851 (Argyros et al.). Methods to enhance the reuse glycerol pathway by over expression of glycerol dehydrogenase (GCY1) and dihydroxyacetone kinase (DAK1) to convert glycerol to dihydroxyacetone phosphate (Zhang et al. (2013) *J. Ind. Microbiol. Biotechnol.* 40:1153-60).

(42) The modified yeast may further feature increased acetyl-CoA synthase (also referred to acetyl-CoA ligase) activity (EC 6.2.1.1) to scavenge (i.e., capture) acetate produced by chemical or enzymatic hydrolysis of acetyl-phosphate (or present in the culture medium of the yeast for any other reason) and converts it to Ac-CoA. This partially reduces the undesirable effect of acetate on the growth of yeast cells and may further contribute to an improvement in alcohol yield. Increasing acetyl-CoA synthase activity may be accomplished by introducing a heterologous acetyl-CoA synthase gene into cells, increasing the expression of an endogenous acetyl-CoA synthase gene and the like.

(43) In some embodiments the modified cells may further include a heterologous gene encoding a protein with NAD^{sup.}-dependent acetylating acetaldehyde dehydrogenase activity and/or a heterologous gene encoding a pyruvate-formate lyase. The introduction of such genes in combination with attenuation of the glycerol pathway is described. e.g., in U.S. Pat. No. 8,795,998 (Pronk et al.). In some embodiments of the present compositions and methods the yeast expressly lacks a heterologous gene(s) encoding an acetylating acetaldehyde dehydrogenase, a pyruvate-formate lyase or both.

(44) In some embodiments, the present modified yeast cells may further over-express a sugar transporter-like (STL) polypeptide to increase the uptake of glycerol (see. e.g., Ferreira et al. (2005) *Mol. Biol. Cell.* 16:2068-76; Dušková et al. (2015) *Mol. Microbiol.* 97:541-59 and WO 2015023989 A1) to increase ethanol production and reduce acetate.

(45) In some embodiments, the present modified yeast cells further include a butanol biosynthetic pathway. In some embodiments, the butanol biosynthetic pathway is an isobutanol biosynthetic pathway. In some embodiments, the isobutanol biosynthetic pathway comprises a polynucleotide

encoding a polypeptide that catalyzes a substrate to product conversion selected from the group consisting of: (a) pyruvate to acetolactate; (b) acetolactate to 2,3-dihydroxyisovalerate; (c) 2,3-dihydroxyisovalerate to 2-ketoisovalerate; (d) 2-ketoisovalerate to isobutyraldehyde; and (e) isobutyraldehyde to isobutanol. In some embodiments, the isobutanol biosynthetic pathway comprises polynucleotides encoding polypeptides having acetolactate synthase, keto acid reductoisomerase, dihydroxy acid dehydratase, ketoisovalerate decarboxylase, and alcohol dehydrogenase activity.

(46) In some embodiments, the modified yeast cells comprising a butanol biosynthetic pathway further comprise a modification in a polynucleotide encoding a polypeptide having pyruvate decarboxylase activity. In some embodiments, the yeast cells comprise a deletion, mutation, and/or substitution in an endogenous polynucleotide encoding a polypeptide having pyruvate decarboxylase activity. In some embodiments, the polypeptide having pyruvate decarboxylase activity is selected from the group consisting of: PDC1, PDC5, PDC6, and combinations thereof. In some embodiments, the yeast cells further comprise a deletion, mutation, and/or substitution in one or more endogenous polynucleotides encoding FRA2, ALD6, ADH1, GPD2, BDH1, DLS1, DPB3, CPR1, MAL23C, MNN4, PAB1, TMN2, HAC1, PTC1, PTC2, OSM1, GIS1, CRZ1, HUG1, GDS1, CYB2P, SFC1, MVB12, LDB10, C5SD, GIC1, GIC2 and/or YMR226C.

V. Combination of Increased Expression β -Glucosidase with Other Beneficial Mutations

(47) In some embodiments, in addition to increased expression of β -glucosidase polypeptides, optionally in combination with other genetic modifications that benefit alcohol production and/or acetate reduction, the present modified yeast cells further include any number of additional genes of interest encoding proteins of interest. Additional genes of interest may be introduced before, during, or after genetic manipulations that result in the increased production of β -glucosidase polypeptides. Proteins of interest, include selectable markers, carbohydrate-processing enzymes, and other commercially-relevant polypeptides, including but not limited to an enzyme selected from the group consisting of a dehydrogenase, a transketolase, a phosphoketolase, a transaldolase, an epimerase, a phytase, a xylanase, a β -glucanase, a phosphatase, a protease, an α -amylase, a β -amylase, a glucoamylase, a pullulanase, an isoamylase, a cellulase, a trehalase, a lipase, a pectinase, a polyesterase, a cutinase, an oxidase, a transferase, a reductase, a hemicellulase, a mannanase, an esterase, an isomerase, a pectinases, a lactase, a peroxidase and a laccase. Proteins of interest may be secreted, glycosylated, and otherwise-modified.

VI. Use of the Modified Yeast for Increased Alcohol Production

(48) The present compositions and methods include methods for increasing alcohol production and/or reducing glycerol production, in fermentation reactions. Such methods are not limited to a particular fermentation process. The present engineered yeast is expected to be a “drop-in” replacement for convention yeast in any alcohol fermentation facility. While primarily intended for fuel alcohol production, the present yeast can also be used for the production of potable alcohol, including wine and beer.

VII. Yeast Cells Suitable for Modification

(49) Yeasts are unicellular eukaryotic microorganisms classified as members of the fungus kingdom and include organisms from the phyla Ascomycota and Basidiomycota. Yeast that can be used for alcohol production include, but are not limited to, *Saccharomyces* spp., including *S. cerevisiae*, as well as *Kluyveromyces*, *Lachancea* and *Schizosaccharomyces* spp. Numerous yeast strains are commercially available, many of which have been selected or genetically engineered for desired characteristics, such as high alcohol production, rapid growth rate, and the like. Some yeasts have been genetically engineered to produce heterologous enzymes, such as glucoamylase or α -amylase.

VII. Substrates and Products

(50) Alcohol production from a number of carbohydrate substrates, including but not limited to corn starch, sugar cane, cassava, and molasses, is well known, as are innumerable variations and

improvements to enzymatic and chemical conditions and mechanical processes. The present compositions and methods are believed to be fully compatible with such substrates and conditions. (51) Alcohol fermentation products include organic compound having a hydroxyl functional group (—OH) is bound to a carbon atom. Exemplary alcohols include but are not limited to methanol, ethanol, n-propanol, isopropanol, n-butanol, isobutanol, n-pentanol, 2-pentanol, isopentanol, and higher alcohols. The most commonly made fuel alcohols are ethanol, and butanol.

(52) These and other aspects and embodiments of the present yeast strains and methods will be apparent to the skilled person in view of the present description. The following examples are intended to further illustrate, but not limit, the compositions and methods.

EXAMPLES

Example 1: Addition of β -Glucosidases to Stillage

(53) Fermentation broth from submerged fermentation cultures expressing eight different glycosyl hydrolase enzymes, including ABG54 β -glucosidase, was prepared for analysis by filtration to remove cell mass using 96-well Millipore filter plates. Filtration was followed by exchange of each enzyme from culture broth into 20 mM sodium acetate buffer, pH 5.0, 0.005% TWEEN®-80, using 96-well ZEBATM Spin desalting plates (ThermoFisher, Cat. No.: 89807).

(54) An Agilent HPLC 1290 INFINITY™ system was used to quantify the protein obtained from each culture using a Waters ACQUITY UPLC® C4BEH 300 column (1.7 μm , 1×50 mm). A six-minute program was used starting with an initial gradient from 5% to 33% acetonitrile (Sigma-Aldrich) in 0.5 minutes, followed by a gradient from 33% to 48% acetonitrile in 4.5 minutes, and then a step gradient to 90% acetonitrile. A protein standard curve based on purified *Trichoderma reesei* Bgl1 (referred to as TrBGL1, below) was used to quantify the enzyme sample preparations. Prior to running enzyme samples on the HPLC system all the proteins were first deglycosylated using endoglycosidase-H.

(55) Whole stillage sampled from a grain ethanol production plant was centrifuged and the resulting supernatant passed through a 0.2 μm membrane filter. A 20% volume of 100 mM sodium acetate buffer, pH 5.0, with 0.005% Tween 80 was added to the whole stillage substrate for enzyme activity assays. Glycosyl hydrolase enzymes were added to the filtered whole stillage supernatant to identify enzymes capable of releasing glucose from the soluble recalcitrant oligosaccharides.

(56) Glycosyl hydrolase reactions with the soluble recalcitrant oligosaccharide substrate were performed in 96-well microtiter plates containing 150 μl of the buffered whole stillage supernatant, a 3 μg protein blend of all eight glycosyl hydrolases, and 6 μg of each individual glycosyl hydrolase combined with the protein blend. Reaction plates were sealed, and incubated at 32° C. for 18 hours with 250 RPM mixing. Immediately following the incubation, the plates were unsealed and 150 μl of 0.1 N sulfuric acid was mixed into the well contents to quench the hydrolysis reactions. The amount of glucose product generated by the enzymatic hydrolysis of recalcitrant oligosaccharides was measured using an ABTS assay for glucose.

(57) An ABTS stock solution containing 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) di-ammonium salt (Sigma Aldrich) at 2.88 mg/ml, horseradish peroxidase (Sigma Aldrich) at 0.11 U/ml and glucose oxidase (OXYGO™ HP 5000L, DuPont) at 1.05 U/ml was prepared in 50 mM sodium phosphate buffer, pH 7. 95 μl of the ABTS stock solution was transferred into the wells of a 96-well microtiter plate, 5 μl of the quenched reaction plates was pipetted into the assay plates containing the ABTS solution. Assay plates were loaded into a microtiter plate reader set for a 3-min kinetic measurement at an absorbance setting of 405 nm with a 9-sec read interval and a 60-second lag. A 5-sec shaking step was used prior to the kinetic measurement for mixing and to eliminate any air bubbles in the assay plate wells. The amount of glucose produced for each sample reaction was calculated using a standard curve generated with glucose standards and the same ABTS assay. The results are shown in Table 1. The enzyme composition enriched for Enzyme H (the β -glucosidase ABG54) produced the most glucose from a stillage substrate.

(58) TABLE-US-00002 TABLE 1 Production of glucose from recalcitrant soluble oligosaccharides

in whole stillage Enzyme Glucose produced (mM) Protein blend (PB) 7.99 PB + Enzyme A 8.37 PB + Enzyme B 8.16 PB + Enzyme C 8.17 PB + Enzyme D 8.15 PB + Enzyme E 8.22 PB + Enzyme F 8.08 PB + Enzyme G 8.10 PB + Enzyme H 9.65

Example 2: Materials and Methods for Examples 2-8

(59) Liquefact Preparation

(60) Liquefact (Ground corn slurry) was prepared by adding 600 ppm of urea, 0.124 SAPU/g ds FERMGEN™ 2.5×(acid fungal protease), 0.33 GAU/g ds variant *Trichoderma* glucoamylase (TrGA) and 1.46 SSCU/g ds *Aspergillus* α-amylase (AkAA), adjusted to a pH of 4.8. For evaluation of strains expressing glucoamylase, the dose of glucoamylase was reduced to 0.1 GAU/g ds. In certain experiments, 5 g/L cellobiose was also added to the liquefact.

(61) Serum Vial Assays

(62) 2 mL of YPD in 24-well plates were inoculated with yeast cells and the cells allowed to grow overnight, 5 ml prepared liquefact was aliquoted to serum vials (Chemglass, Catalog No.: CG-4904-01) and yeast was added to each vial for a final OD of about 0.2-0.4. The lids of the vials were screw on and punctured with a needle (BD, Catalog No.: 305111) for ventilation (to release CO₂), then incubated at 32° C. with shaking (200 RPM) for 55 hours.

(63) Shake Flask Assays

(64) 100 µL of concentrated yeast overnight culture was added to each of a number of shake flasks filled with 50 g prepared liquefact for a final OD of 0.3. The flasks were incubated at 32° C. with shaking (200 RPM) for 55 hours.

(65) HPLC Analysis

(66) Samples from serum vial and shake flask experiments were filtered through 0.2 µm PTFE filters and the filtrates analyzed for acetate, ethanol, glycerol, glucose, DP2, DP3 and DP4+ content by HPLC (Waters e2695 series) using Bio-Rad Aminex HPX-87H columns at 65° C., with an isocratic flow rate of 0.6 ml/min in 0.01 N H₂SO₄ eluent. A 2.5 µl sample injection volume was used. Calibration standards were used for quantification of acetate, ethanol, glycerol and glucose. Values are expressed in g/L.

(67) Growth of Strains for β-Glucosidase Activity Determination

(68) Yeast strains were inoculated into 2 ml synthetic complete medium lacking uracil (SC-ura) or YPD medium in 24-well plates and the cells allowed to grow overnight. Cultures were filtered through a 0.2 µm filter plate (PALL AcroPrep Advance, GHP membrane) and the resulting filtrates assayed for β-glucosidase activity.

(69) β-Glucosidase Activity Assay

(70) A 4-nitrophenyl-β-D-glucopyranoside (pNPG) substrate solution was prepared by dissolving 0.006 g pNPG in 20 ml 0.05 M sodium acetate buffer (0.05 M sodium acetate, 0.1% (v/v) Polyethylene glycerol 8000, pH 4.8). 100 µl pNPG substrate solution was mixed with 20 µl filtered culture supernatant or β-glucosidase standard and incubated for 45 min at 30°C. The reaction was stopped by adding 50 µl 1 M sodium carbonate and incubating for 1 min at room temperature, then absorbance at 405 nm was measured. β-glucosidase activity in culture supernatants was calculated by comparison to a standard curve generated using serially diluted β-glucosidase standard.

(71) Signal Sequence Prediction

(72) The determination of signal sequence cleavage sites was predicted using SignalP version 4.1 (Petersen, T. N. et al. (2011) Nature Methods, 8:785-86) with default parameters. It will be understood that actual signal sequence cleavage sites, and therefore the N-termini of mature secreted polypeptides, may differ by a few amino acid residues or may vary within a population of secreted polypeptides.

Example 3: Screening for Expression of β-Glucosidases in *S. cerevisiae*

(73) Genes coding for four different β-glucosidases (SEQ ID NOs: 1-4), codon optimized for *Saccharomyces cerevisiae*, were cloned between the SpeI and NotI restriction sites in pJT257 (described in U.S. Pat. No. 9,181,566). In the resulting plasmids, expression of the β-glucosidase is

under control of the FBA1 promoter and FBA1 terminator from *S. cerevisiae* (see, e.g., WO2018/111792). A second set of plasmids was constructed containing genes coding for the same four β -glucosidases but with the native signal sequence (SEQ ID NOs: 5-8) removed and replaced with the signal sequence from the *S. cerevisiae* mating factor α (MFalpha; SEQ ID NO: 9). Plasmids were designated as show in Table 2. Note that FAB is hybrid enzyme derived from portions of the β -glucosidases from *Fusarium vesticilloies*, *Rasamsonia emersonii* and *Hypocrea* (*Trichoderma*) *jecorina*, which is described in WO2012125951 (Kaper et al.) as SEQ ID NO: 135.

(74) TABLE-US-00003 TABLE 2 β -glucosidase expression plasmids

Plasmid	β -glucosidase	Signal sequence	SEQ ID NO	Source organism
pYKH1127	ABG54	native	5	<i>Gomerella graninicola</i>
pYKH1139	MFalpha	9		<i>Fusarium vesticilloies</i>
pYKH1135	FAB	2		<i>Fusarium vesticilloies</i>
pYKH1095	MFalpha	9		<i>Rasamsonia emersonii</i>
pYKH1097	Mg3A	3		<i>Hypocrea jecorina</i>
pYKH1096	MFalpha	9		<i>Magnaporthe grisea</i>
pYKH1099	TrBGL1	4		<i>Hypocrea jecorina</i>
pYKH1098	MFalpha	9		<i>Hypocrea jecorina</i>

(75) The plasmids listed in Table 2 were transformed into strain FG-ura3 and transformants were selected by growth on synthetic complete medium lacking uracil (SC-ura). FG-ura3 is a derivative of FERMAX™ Gold (hereafter abbreviated, “FG”) in which the ura3 gene has been deleted. The construction of FG-ara3 is described in WO2018111792A1. FG-ura3 was also transformed with the control plasmid pPOL00040 that lacks a β -glucosidase coding sequence. Transformants were grown overnight in SC-ura liquid medium, and β -glucosidase activity in the culture medium was measured as described in Example 2. The results from two such experiments are shown in Table 3.

(76) TABLE-US-00004 TABLE 3 β -glucosidase activity measured in FG-ura3 transformed with β -glucosidase expression plasmids

Signal	Plasmid	Enzyme sequence	A.sub.405 (Exp. 1)	A.sub.405 (Exp. 2)
pYKH1127	ABG54	native	0.107 ± 0.018	0.174 ± 0.023
pYKH1139	MFalpha	n/d	0.244 ± 0.055	0.152 ± 0.014
pYKH1135	FAB	native	0.086 ± 0.004	0.156 ± 0.009
pYKH1095	MFalpha	0.152 ± 0.014	0.308 ± 0.057	0.005 ± 0.001
pYKH1097	Mg3A	native	0.005 ± 0.001	n/d
pYKH1096	MFalpha	0.004 ± 0.001	n/d	0.004 ± 0.001
pYKH1099	TrBGL1	native	0.004 ± 0.001	n/d
pYKH1098	MFalpha	0.004 ± 0.000	n/d	0.004 ± 0.000
pPOL00040	none	n/a	0.003 ± 0.001	0.003 ± 0.001

(77) A.sub.405 was higher for strains containing plasmids encoding ABG54 or FAB than for the control strain transformed with pPOL00040. This demonstrates that strains encoding ABG54 or FAB express β -glucosidase activity. β -glucosidase activity was detected for strains expressing ABG54 or FAB with either their native signal sequences or the MFalpha signal sequence. In contrast, strains containing plasmids encoding Mg3A or TrBGL1 did not appear to produce β -glucosidase activity above that measured for the negative control strain transformed with pPOL00040.

Example 4: Strains with Integrated Expression Cassettes for ABG54 or FAB

(78) Expression cassettes coding for ABG54 with its native signal sequence, ABG54 with the MFalpha signal sequence, FAB with its native signal sequence (SEQ ID NO:6) or FAB with the MFalpha signal sequence (SEQ ID NO:9) were amplified by PCR from plasmids pYKH1127, pYKH1139, pYKH1135 or pYKH1095, respectively, using primers that incorporate regions of homology to a target site at the PAM1 locus in *S. cerevisiae*. Each of the amplified DNA fragments was used as a donor DNA for CRISPR-mediated integration at the PAM1 locus in FG. Integration of the β -glucosidase expression cassettes was confirmed by colony PCR, and the resulting strains were designated as shown in Table 4.

(79) Strains were also constructed in which the native signal sequence of ABG54 or FAB was replaced with the signal sequence from the *S. cerevisiae* invertase gene, SUC2 (SEQ ID NO: 10). These strains were constructed by transforming FG with two overlapping DNA fragments—one containing the *S. cerevisiae* FBA1 promoter joined to the SUC2 signal sequence, and the other containing the ABG54 or FAB coding sequence (without the native signal sequence) joined to the FBA1 terminator. These fragments were assembled and integrated at the PAM1 locus using the procedure described in EauClaire et. al. (2016) *J. Ind. Microbiol. Biotechnol.* 43:1001-15. Correct

assembly and integration of the β -glucosidase expression cassettes was confirmed by colony PCR. The resulting strains were designated as shown in Table 4.

(80) Four clones of each strain, together with the parental strain FG, were grown overnight in YPD liquid medium, and β -glucosidase activity in the resulting culture medium was measured as described in Example 2. The results are shown in Table 4.

(81) TABLE-US-00005 TABLE 4 β -glucosidase activity secreted by strains encoding ABG54 or FAB, versus FG. β -glucosidase Strain expressed Signal sequence A.sub.405 GKH-0464 ABG54 native 0.970 ± 0.031 GKH-0459 MFalpha 1.240 ± 0.107 GKH-0484 SUC2 0.922 ± 0.016 GKH-0455 FAB native 0.248 ± 0.011 GKH-0450 MFalpha 0.666 ± 0.039 GKH-0466 SUC2 0.170 ± 0.053 FG n/a n/a 0.088 ± 0.012

(82) A.sub.405 was higher for the new strains with integrated β -glucosidase expression cassettes than for the parental FG strain. This indicates that strains GKH-0464, GKH-0459, GKH-0484, GKH-0455, GKH-0450 and GKH-0466 all express β -glucosidase activity.

Example 5. Ethanol Production by *S. cerevisiae* Strains Expressing β -Glucosidase in Vial Assays

(83) Four clones of each strain listed in Table 4 were screened for ethanol production, relative to the parent strain FG. Strains were growth in corn liquefact in serum vials, and ethanol production was analyzed after 55 h fermentation as described in Example 2. The results are shown in Table 5.

(84) TABLE-US-00006 TABLE 5 Performance of β -glucosidase expressing yeast strains versus FG in vial assays Ethanol Signal increase Strain β -glucosidase sequence Ethanol (g/L) vs FG (%)
GKH-0464 ABG54 native 142.41 ± 0.31 1.5 GKH-0459 MFalpha 141.35 ± 0.59 0.7 GKH-0484 SUC2 142.02 ± 0.92 1.2 GKH-0455 FAB native 141.99 ± 0.12 1.2 GKH-0450 MFalpha 138.94 ± 0.41 -1.0 GKH-0466 SUC2 142.31 ± 0.33 1.4 FG n/a n/a 140.37 ± 0.40 n/a

(85) The (3-glucosidase expressing strains GKH-0464, GKH-0459, GKH-0484, GKH-0455 and GKH-0466 resulted in up to 1.5% increased ethanol production compared to the parental FG strain.

Example 6: Ethanol Production by *S. cerevisiae* Strains Expressing β -Glucosidase in Shake Flask Assays

(86) To further examine the benefits of β -glucosidase expression, the performance of strains GKH-0464 and GKH-0484 were analyzed more precisely using a shake flask assay. Strains GKH-0464, GKH-0484 or the parental strain FG were grown for 55 h in corn liquefact, and their fermentation products analyzed as described in Example 2. In a second set of flasks, approx. 5 g/L cellobiose was added prior to inoculation with the yeast. The results are shown in Table 6.

(87) TABLE-US-00007 TABLE 6 Performance of β -glucosidase expressing yeast strains versus FG in shake flask assays with or without cellobiose ABG54 Cellobiose Ethanol Glycerol Acetate
Glucose DP2 DP3 Strain exp. added (g/L) (g/L) (g/L) (g/L) (g/L) (g/L) GKH-0464 Yes No 146.16 16.01 0.88 2.12 4.15 1.25 GKH-0484 Yes No 143.87 16.36 0.88 4.87 4.12 1.40 FG No No 142.02 16.25 0.92 5.73 4.24 1.58 GKH-0464 Yes Yes 147.00 16.34 0.90 2.86 4.31 1.37 GKH-0484 Yes Yes 143.36 16.39 0.92 7.58 4.64 1.50 FG No Yes 142.25 16.19 0.96 6.90 10.16 1.77

(88) The β -glucosidase-expressing strains GKH-0464 and GKH-0484 resulted in up to 3.3% increased ethanol production compared to the parental FG strain. GKH-0464 and GKH-0484 also resulted in up to 6.3% decreased acetate, as well as decreased DP2 and decreased DP3 compared to the parental strain. For fermentations with FG, addition of approximately 5 g/L cellobiose to the liquefact prior to inoculation with the yeast resulted in a corresponding increase in DP2 at the end of fermentation. This indicates that the added cellobiose was not consumed during fermentation with FG. In contrast, for GKH-0464 and GKH-0484, the DP2 concentration at the end of fermentation was similar for liquefact with or without added cellobiose. This indicates that cellobiose can be consumed during fermentation with GKH-0464 and GKH-0484.

Example 7: Strains Expressing β -Glucosidase with the PKL Pathway and Glucoamylase

(89) An ABG54 coding sequence, identical to that in plasmid pYKH1127, flanked by the *S. cerevisiae* FBA1 promoter and GPD1 terminator, was integrated at the JIP5 locus in *S. cerevisiae*. The amplified DNA fragment was used as a donor DNA for CRISPR-mediated integration at the

JIP5 locus in two parental strains: (i) FG-PKL and (ii) FG-PKL-GA. FG-PKL is an engineered FG yeast having a heterologous phosphoketolase (PKL) pathway involving the expression of phosphoketolase (PKL), phosphotransacetylase (PTA) and acetylating acetyl dehydrogenase (AADH), as described in WO2015148272. FG-PKL-GA is the FG-PKL strain further engineered to expresses a variant of *Trichoderma* glucoamylase.

(90) Integration of the ABG54 expression cassette into these strains was confirmed by colony PCR. The resulting strains, along with the parental strains lacking the ABG54 expression cassette, were grown in corn liquefact in serum vials, and their fermentation products analyzed as described in Example 2. The results are shown in Table 7.

(91) TABLE-US-00008 TABLE 7 Performance of β -glucosidase expressing strains G3020 and G3014 versus parental strains in vial assays

Strains	Ethanol (g/L)	Glycerol (g/L)	Acetate (g/L)	Glucose DP2 (g/L)	Glucose DP3 (g/L)
FG-PKL PKL pathway	145.27	12.84	1.77	2.17	4.12
G3020 PKL pathway + ABG54	145.87	12.61	1.69	1.48	3.18
FG-PKL-GA PKL pathway + ABG54	146.73	11.64	1.64	1.49	4.27
G3014 PKL pathway + ABG54	147.40	11.75	1.43	1.46	3.61

(92) The β -glucosidase-expressing strains G3020 and G3014 resulted in slightly increased ethanol production (<0.5%) compared to their corresponding parental strains. However, G3020 and G3014 resulted in up to almost 13% decreased acetate, as well as decreased DP2 and decreased DP3 compared to their respective parental strains.

Example 8: Hybrid Yeast Strains Expressing β -Glucosidase

(93) The ABG54 expression cassette from plasmid pYKH1127 was amplified by PCR using primers that incorporate regions of homology to a target site at the JEN1 locus in *S. cerevisiae*. The amplified DNA fragment was used as a donor DNA for CRISPR-mediated integration at the JEN1 locus in two parental strains: (i) DGY1- Δ and (ii) DGY1- Δ -GA. DGY1- Δ is a hybrid yeast strain produced by mating two commercially-available parental yeast strains and further modified by deletion of the YJL065c gene encoding Dls1 (see, e.g., WO2018089333). DGY1- Δ -GA is the DGY1- Δ strain further modified to expresses a glucoamylase.

(94) Integration of the ABG54 expression cassette into these strains was confirmed by colony PCR. The resulting strains, along with the parental strains lacking the ABG54 expression cassette, were grown in corn liquefact in shake flasks, and their fermentation products analyzed as described in Example 2. The results are shown in Table 8.

(95) TABLE-US-00009 TABLE 8 Performance of β -glucosidase-expressing hybrid strains versus parental strains

Strains	Ethanol (g/L)	Glycerol (g/L)	Acetate (g/L)	Glucose DP2 (g/L)	Glucose DP3 (g/L)
DGY1- Δ Hybrid	143.64	13.41	1.62	0.23	2.89
GKH-0737 Hybrid + ABG54	144.72	13.47	1.58	0.33	2.72
DGY1- Δ -GA Hybrid + ABG54	144.17	12.69	1.36	0.34	3.15
GKH-0732 Hybrid + ABG54	145.46	12.68	1.26	0.18	4.02

(96) The β -glucosidase-expressing strains GKH-0737 and GKH-0732 resulted in up to about 0.9% increased ethanol production compared to their corresponding parental strains and up to over 7% decreased acetate, as well as decreased DP3, compared to their respective parental strains.

Example 9: Comparison of ABG54 and FAB with Other β -Glucosidases In Vitro

(97) Following the observation that ABG54 and FAB were the best expressed β -glucosidases in yeast (Example 3), an experiment similar to that described in Example 1 was performed to compare the activity of ABG54 and FAB to other β -glucosidases in the aforementioned whole stillage assay.

(98) As shown in Example 1, ABG54 was the best performing glycosyl hydrolase for releasing glucose from soluble recalcitrant oligosaccharides in whole stillage, leading to the impetus to express β -glucosidases in yeast. As shown in Example 3, ABG54 and FAB were shown to be the best expressing β -glucosidases for expression in yeast.

(99) In an essentially a fill-circle experiment summarized by the results shown in Table 9, ABG54 and FAB are demonstrated to be better performing β -glucosidases than both Mg3A and TrBg11 (the two other β -glucosidases tested in yeast) and superior to a blend of other GH-15 and GH-3

molecules. Notably and not surprisingly, all the enzymes selected and tested, herein, were considered best-in-class, otherwise they would not have been tested.

(100) Regardless, the measurable superior performance of ABG54 and FAB compared to Mg3A and TrBgl1 (in particular) in no way explains the remarkable difference in expression in yeast. The differences in expression of the β -glucosidases are ordinals of magnitude greater than the difference in performance of the enzymes in vitro. These data confirm the tenet that expression of exogenous enzymes in yeast is often problematic, and that ABG54 and FAB are especially well expressed, and are additionally superior within their class. These results are shown in Table 9.

(101) TABLE-US-00010 TABLE 9 Comparison of in vitro performance of β -glucosidases expressed in yeast Enzyme combination Glucose produced (mM) PB + ABG54 5.99 PB + FAB 5.61 PB + Mg3A 4.92 PB + TrBgl1 5.49 GH15 + GH3 blend 3.61

Claims

1. Modified yeast cells derived from parental yeast cells, the modified yeast cells comprising a genetic alteration that causes the modified yeast cells to produce an increased amount of β -glucosidase polypeptides compared to the parental yeast cells, wherein the modified yeast cells produce during fermentation more ethanol and/or less acetate compared to the amount of ethanol and acetate produced by otherwise identical parental yeast cells, wherein the genetic alteration comprises introduction of an expression cassette for expressing a β -glucosidase polypeptide, and wherein the β -glucosidase polypeptides have: a) an amino acid sequence having at least 80% amino acid sequence identity to SEQ ID NO: 1 or SEQ ID NO: 2; b) the amino acid sequence of a mature polypeptide encoded by a nucleic acid having at least 80% nucleic acid sequence identity to any of SEQ ID NOs: 11-16; or c) the amino acid sequence of a mature polypeptide encoded by a nucleic acid that hybridizes, under stringent hybridization conditions comprising 65° C. and 0.1×SSC, to any of SEQ ID NOs: 11-16, or the complement, thereof.
2. The modified yeast cells of claim 1, further comprising an exogenous gene encoding a carbohydrate processing enzyme, wherein the carbohydrate processing enzyme is a dehydrogenase, a transketolase, a phosphoketolase, a transaldolase, an epimerase, a phytase, a xylanase, a β -glucanase, a phosphatase, a protease, an α -amylase, a β -amylase, a glucoamylase, a pullulanase, an isoamylase, a cellulase, a trehalase, a lipase, a pectinase, a polyesterase, a cutinase, an oxidase, a transferase, a reductase, a hemicellulase, a mannanase, an esterase, an isomerase, a pectinases, a lactase, a peroxidase, or a laccase.
3. The modified yeast cells of claim 1, further comprising an exogenous PKL pathway.
4. The modified yeast cells of claim 1, further comprising a reduction or elimination of an endogenous NAD-dependent 3-phosphate dehydrogenase (GPD); a disruption of one or more of genes GPD1, GPD2, GPP1, and/or GPP2; overexpression of glycerol dehydrogenase (GCY1); and/or overexpression of dihydroxyacetone kinase (DAK1).
5. The modified yeast cells of claim 1, wherein the modified yeast cells further make reduced amounts of DP2 and/or DP3 compared to otherwise identical parental yeast cells.
6. The modified yeast cells of claim 1, wherein the β -glucosidase polypeptides are derived from *Glomerella graminicola*.
7. The modified yeast cells of claim 1, wherein the cells are of a *Saccharomyces* spp.
8. A method for increasing the production of alcohol and/or decreasing the production of acetate from yeast cells grown on a carbohydrate substrate, comprising: introducing into parental yeast cells a genetic alteration that increases the production of β -glucosidase polypeptides compared to the amount produced in the parental yeast cells, wherein the cells having the introduced genetic alteration are the modified yeast cells of claim 1.
9. The method of claim 8, wherein the increased production of alcohol is at least 0.5%, at least 1.0%, at least 2.0% or at least 3.0%.

10. The method of claim 8, wherein the decreased production of acetate is at least 1.0%, at least 2.0%, at least 4.0% or at least 6.0%.
 11. The method of claim 8, wherein the modified yeast cells further comprise an exogenous PKL pathway.
 12. The modified yeast cells of claim 1, further comprising a heterologous acetyl-CoA synthase gene and/or increased expression of an endogenous acetyl-CoA synthase gene.
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